

Assignment_Vector Spaces

1. A real-valued function f defined on the real line is called an **even function** if $f(-t) = f(t)$ for each real number t . Prove that the set of even functions defined on the real line with the operations of addition $((f+g)(x) = f(x) + g(x))$ and scalar multiplication $((af)(x) = af(x))$ is a vector space.

2. Let V and W be vector spaces over a field F . Let

$$Z = \{(v, w) : v \in V \text{ and } w \in W\}.$$

Prove that Z is a vector space over F with the operations

$$(v_1, w_1) + (v_2, w_2) = (v_1 + v_2, w_1 + w_2) \text{ and } c(v_1, w_1) = (cv_1, cw_1).$$

3. Let S be a nonempty set and F a field. Let $C(S, F)$ denote the set of all functions $f \in F(S, F)$ such that $f(s) = 0$ for all but a finite number of elements of S . Prove that $C(S, F)$ is a subspace of $F(S, F)$.

4. Let W_1 and W_2 be subspaces of a vector space V . Prove that $W_1 \cup W_2$ is a subspace of V if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

5. Let W be a subspace of a vector space V over a field F . For any $v \in V$ the set $\{v\} + W = \{v + w : w \in W\}$ is called the **coset** of W **containing** v . It is customary to denote this coset by $v + W$ rather than $\{v\} + W$.

(a) Prove that $v + W$ is a subspace of V if and only if $v \in W$.

(b) Prove that $v_1 + W = v_2 + W$ if and only if $v_1 - v_2 \in W$.

Addition and scalar multiplication by scalars of F can be defined in the collection $S = \{v + W : v \in V\}$ of all cosets of W as follows:

$$(v_1 + W) + (v_2 + W) = (v_1 + v_2) + W$$

for all $v_1, v_2 \in V$ and

$$a(v + W) = av + W, \text{ for all } v \in V \text{ and } a \in F.$$

(c) Prove that the preceding operations are well defined; that is, show that if $v_1 + W = v'_1 + W$ and $v_2 + W = v'_2 + W$, then

$$(v_1 + W) + (v_2 + W) = (v'_1 + W) + (v'_2 + W)$$

$$\text{And } a(v_1 + W) = a(v'_1 + W)$$

for all $a \in F$.

(d) Prove that the set S is a vector space with the operations defined in

(c). This vector space is called the **quotient space of V modulo W** and is denoted by V/W .

6. Determine whether the given vector is in the span of S .

$$2x^3 - x^2 + x + 3, S = \{x^3 + x^2 + x + 1, x^2 + x + 1, x + 1\}$$

7. Show that a subset W of a vector space V is a subspace of V if and only if $\text{span}(W) = W$.

8. Show that if S_1 and S_2 are arbitrary subsets of a vector space V , then $\text{span}(S_1 \cup S_2) = \text{span}(S_1) + \text{span}(S_2)$.

9. Let S_1 and S_2 be subsets of a vector space V . Prove that $\text{span}(S_1 \cap S_2) \subseteq \text{span}(S_1) \cap \text{span}(S_2)$. Give an example in which $\text{span}(S_1 \cap S_2)$ and $\text{span}(S_1) \cap \text{span}(S_2)$ are equal and one in which they are unequal.

10. Determine whether the following sets are linearly dependent or linearly independent.

(i) $\{x^3 - x, 2x^2 + 4, -2x^3 + 3x^2 + 2x + 6\}$ in $P_3(R)$ (polynomials with degree ≤ 3)

(ii) $\{(1, -1, 2), (1, -2, 1), (1, 1, 4)\}$ in R^3

11. Let $S = \{u_1, u_2, \dots, u_n\}$ be a finite set of vectors. Prove that S is linearly dependent if and only if $u_1 = 0$ or $u_{k+1} \in \text{span}(\{u_1, u_2, \dots, u_k\})$ for some k ($1 \leq k < n$).

12. Determine which of the following sets are bases for R^3 .

(i) $\{(-1, 3, 1), (2, -4, -3), (-3, 8, 2)\}$

(ii) $\{(1, -3, -2), (-3, 1, 3), (-2, -10, -2)\}$

13. The vectors $u_1 = (2, -3, 1)$, $u_2 = (1, 4, -2)$, $u_3 = (-8, 12, -4)$, $u_4 = (1, 37, -17)$, and $u_5 = (-3, -5, 8)$ generate R^3 .

Find a subset of the set

$\{u_1, u_2, u_3, u_4, u_5\}$ that is a basis for R^3 . Find the coordinates of the vector $v = (1, 1, -1)$ w.r.t. this basis.

14. Let u, v , and w be distinct vectors of a vector space V . Show that if $\{u, v, w\}$ is a basis for V , then $\{u+v+w, v+w, w\}$ is also a basis for V .

15. Given $A = \begin{pmatrix} 2 & -1 & 3 & 2 \\ 1 & 4 & 0 & -1 \\ 2 & 6 & -1 & 5 \end{pmatrix}$. (i) Find a basis of $\text{Nul}(A)$, (ii) find a basis for $\text{Row}(A)$, (iii) find a basis for $\text{Col}(A)$, (iv) find a matrix B such that $\text{Nul}(B) = \text{Row}(A)$.

16. Find bases for the following subspaces of $\mathbb{R}^5$:

$$W_1 = \{(a_1, a_2, a_3, a_4, a_5) \in \mathbb{R}^5 : a_1 - a_3 - a_4 = 0\},$$

$$W_2 = \{(a_1, a_2, a_3, a_4, a_5) \in \mathbb{R}^5 : a_2 = a_3 = a_4 \text{ and } a_1 + a_5 = 0\}, \quad W = W_1 + W_2,$$

$$W_1 \cap W_2.$$

What are the dimensions of W_1 , W_2 , W , $W_1 \cap W_2$?

17. Let W_1 and W_2 be subspaces of a finite-dimensional vector space V . Determine necessary and sufficient conditions on W_1 and W_2 so that $\dim(W_1 \cap W_2) = \dim(W_1)$.

18. Let $v_1, v_2, \dots, v_k, v$ be vectors in a vector space V , and define

$$W_1 = \text{span}(\{v_1, v_2, \dots, v_k\}), \text{ and } W_2 = \text{span}(\{v_1, v_2, \dots, v_k, v\}).$$

(a) Find necessary and sufficient conditions on v such that $\dim(W_1) = \dim(W_2)$.

(b) State and prove a relationship involving $\dim(W_1)$ and $\dim(W_2)$ in the case that $\dim(W_1) \neq \dim(W_2)$.

19. Let W_1 and W_2 be subspaces of a vector space V having dimensions m and n , respectively, where $m \geq n$.

(a) Prove that $\dim(W_1 \cap W_2) \leq n$.

(b) Prove that $\dim(W_1 + W_2) \leq m + n$.

20. (a) Let W_1 and W_2 be subspaces of a vector space V such that $V = W_1 \oplus W_2$. If β_1 and β_2 are bases for W_1 and W_2 , respectively, show that $\beta_1 \cap \beta_2 = \emptyset$ and $\beta_1 \cup \beta_2$ is a basis for V .

(b) Conversely, let β_1 and β_2 be disjoint bases for subspaces W_1 and W_2 , respectively, of a vector space V . Prove that if $\beta_1 \cup \beta_2$ is a basis for V , then $V = W_1 \oplus W_2$.